$\mathcal{K}_{0^+}^{\#1}$ $\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$

$-\frac{3\,Y_1\,k^2}{8}$	0
0	0

$\mathcal{K}_{0^+}^{\#1\,\dagger}$, $\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$

$\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$ $\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$ $\mathcal{K}_{1^-}^{\#1\,\alpha}$ $\mathcal{K}_{1^-}^{\#2\,\alpha}$

$-\frac{Y_2\,k^2}{8}$	$\frac{Y_2\,k^2}{4\sqrt{2}}$	0	0
$\frac{Y_2\,k^2}{4\sqrt{2}}$	$-\frac{Y_2\,k^2}{4}$	0	0
0	0	$-k^2\,{}^{(4)}\kappa_2$	$-\sqrt{2}\,k^2\,{}^{(4)}\kappa_2$
0	0	$-\sqrt{2}\,k^2\,{}^{(4)}\kappa_2$	$-2\,k^2\,{}^{(4)}\kappa_2$

$\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta}$, $\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta}$, $\mathcal{K}_{1^-}^{\#1\,\dagger\alpha}$, $\mathcal{K}_{1^-}^{\#2\,\dagger\alpha}$

$\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$ $\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$

$\frac{9\,Y_3\,k^2}{8}$	0
0	0

$\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta}$, $\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$

$\mathcal{J}_{0^+}^{\#1}$ $\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$

$\frac{1}{\text{Det}\,(0^+)}$	0
0	0

$\mathcal{J}_{0^+}^{\#1\,\dagger}$, $\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$

$\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$ $\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$ $\mathcal{J}_{1^-}^{\#1\,\alpha}$ $\mathcal{J}_{1^-}^{\#2\,\alpha}$

$\frac{1}{3\,\text{Det}\,(1^+)}$	$-\frac{\sqrt{2}}{3\,\text{Det}\,(1^+)}$	0	0
$-\frac{\sqrt{2}}{3\,\text{Det}\,(1^+)}$	$\frac{2}{3\,\text{Det}\,(1^+)}$	0	0
0	0	$-\frac{1}{9\,k^2\,{}^{(4)}\kappa_2}$	$-\frac{\sqrt{2}}{9\,k^2\,{}^{(4)}\kappa_2}$
0	0	$-\frac{\sqrt{2}}{9\,k^2\,{}^{(4)}\kappa_2}$	$-\frac{2}{9\,k^2\,{}^{(4)}\kappa_2}$

$\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta}$, $\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta}$, $\mathcal{J}_{1^-}^{\#1\,\dagger\alpha}$, $\mathcal{J}_{1^-}^{\#2\,\dagger\alpha}$

$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$ $\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$

$\frac{1}{\text{Det}\,(2^+)}$	0
0	0

$\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta}$, $\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$

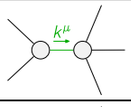
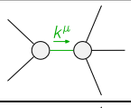
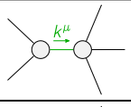
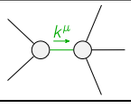
Abbreviations used in matrices

$$Y_1 == 4\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 + 3\,{}^{(4)}\kappa_3 \;\&\&\; Y_2 == 9\,{}^{(4)}\kappa_2 - {}^{(4)}\kappa_3 \;\&\&\; Y_3 == {}^{(4)}\kappa_2 - {}^{(4)}\kappa_3 \;\&\&$$
$$\text{Det}\,(0^+) == -\frac{3}{8}\,k^2\,(4\,{}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 + 3\,{}^{(4)}\kappa_3) \;\&\&\; \text{Det}\,(1^+) == \frac{3}{8}\,k^2\,(-9\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_3) \;\&\&\; \text{Det}\,(2^+) == \frac{9}{8}\,k^2\,({}^{(4)}\kappa_2 - {}^{(4)}\kappa_3)$$

Lagrangian

$$\begin{aligned}
&{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\,\partial_\chi\mathcal{K}^{\delta}{}_{\beta\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + \\
&\frac{9}{2}\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} - \frac{9}{4}\,{}^{(4)}\kappa_2\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \\
&\frac{5}{4}\,{}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - 6\,{}^{(4)}\kappa_2\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_\delta\mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{4}\,{}^{(4)}\kappa_2\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{4}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{1^-}^{\#1\,\alpha} == \mathcal{J}_{1^-}^{\#2\,\alpha}$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$2\,\mathcal{J}_{1^+}^{\#1\,\alpha\beta} + \mathcal{J}_{1^+}^{\#2\,\alpha\beta} == 0$	3	$\partial_\chi\mathcal{J}^{\alpha\beta\chi} + \partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\chi\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	12	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{57\,{}^{(4)}\kappa_2^{\,2} - 10\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_3^{\,2}}{9\,{}^{(4)}\kappa_2^{\,3} - 10\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_2\,{}^{(4)}\kappa_3^{\,2}}$
	1	0	$\frac{97\,{}^{(4)}\kappa_2^{\,2} - 58\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 9\,{}^{(4)}\kappa_3^{\,2}}{9\,{}^{(4)}\kappa_2^{\,3} - 10\,{}^{(4)}\kappa_2^{\,2}\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_2\,{}^{(4)}\kappa_3^{\,2}}$
	1	0	$-\frac{216\,{}^{(4)}\kappa_2^{\,2} - 24\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + \sqrt{(-9\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_3)^2\,(1841\,{}^{(4)}\kappa_2^{\,2} - 450\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 81\,{}^{(4)}\kappa_3^{\,2})}}{{}^{(4)}\kappa_2\,(9\,{}^{(4)}\kappa_2^{\,2} - 10\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_3^{\,2})}$
	1	0	$\frac{-216\,{}^{(4)}\kappa_2^{\,2} + 24\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + \sqrt{(-9\,{}^{(4)}\kappa_2 + {}^{(4)}\kappa_3)^2\,(1841\,{}^{(4)}\kappa_2^{\,2} - 450\,{}^{(4)}\kappa_2\,{}^{(4)}\kappa_3 + 81\,{}^{(4)}\kappa_3^{\,2})}}{{}^{(4)}\kappa_2\,({}^{(4)}\kappa_2 - {}^{(4)}\kappa_3)\,(9\,{}^{(4)}\kappa_2 - {}^{(4)}\kappa_3)}$
Resolved unitarity condition(s):	(Demonstrably impossible)		